

A PROJECT REPORT

on

“AI-Driven Renewable Energy Source Prediction and Recommendation Using Ensemble Learning and Renewable Energy Reliability Index”

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In Partial Fulfilment of the Requirement for the Award of

**BACHELOR’S DEGREE IN
COMPUTER SCIENCE ENGINEERING**

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CERTIFICATE

This is certify that the project entitled

**“AI-Driven Renewable Energy Source Prediction and
Recommendation Using Ensemble Learning and Renewable Energy
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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Science & Engineering OR Information Technology) at KIIT Deemed to be university, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

Date: / /

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ABSTRACT

The increasing demand for electricity and the environmental impact of fossil fuels have intensified the need for sustainable and eco-friendly energy solutions. Renewable energy resources such as solar, wind, and hydro power offer clean alternatives; however, identifying the most suitable energy source for a specific region requires accurate analysis of climatic conditions and resource availability. This project presents an AI-driven renewable energy prediction and recommendation system that analyzes environmental parameters to determine the most reliable renewable energy source for electricity generation. The proposed system utilizes real-world climate data obtained from the NASA POWER API, including solar irradiance, wind speed, rainfall, and temperature for Bhubaneswar. The collected data is processed to estimate solar, wind, and hydro energy potentials. Multiple machine learning models, including Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks, are employed to forecast renewable energy potential using historical climate patterns. An ensemble prediction approach combines the outputs of these models to improve forecasting accuracy. To evaluate the stability and reliability of different renewable sources, a novel metric called the Renewable Energy Reliability Index (RERI) is introduced. This index measures the consistency of energy generation potential over time. The model performance is assessed using actual versus predicted comparisons and statistical metrics such as the coefficient of determination (R^2). Experimental results demonstrate high prediction accuracy, with ensemble models achieving performance above 0.99. Based on the calculated energy potentials and reliability index values, the system recommends the most suitable renewable energy source for the target region. The results indicate that solar energy is the most reliable renewable resource for Bhubaneswar. The proposed framework can assist policymakers, energy planners, and researchers in selecting optimal renewable energy solutions for sustainable electricity generation and reducing dependence on fossil fuels.

Keywords:

Artificial Intelligence (AI), Renewable Energy Prediction, Ensemble Machine Learning, Solar Energy Forecasting, Renewable Energy Reliability Index (RERI), Climate Data Analysis, Renewable Energy Recommendation System, Sustainable Energy Planning.

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Chapter 1

Introduction

The rapid growth in global energy demand has created significant challenges for sustainable power generation. Conventional energy sources such as coal, oil, and natural gas have been the primary contributors to electricity production for many decades. However, these fossil fuels are associated with serious environmental problems including air pollution, greenhouse gas emissions, and climate change. As a result, there has been a growing global emphasis on transitioning toward renewable and eco-friendly sources of energy. Renewable energy resources such as solar, wind, and hydro power provide clean and sustainable alternatives that can significantly reduce environmental impact while ensuring long-term energy security.

Among the various renewable sources, solar energy has gained widespread attention due to the abundant availability of sunlight in many regions of the world. Similarly, wind energy has become an important contributor to electricity generation in areas with strong and consistent wind patterns. Hydropower, which relies on flowing or stored water, is another widely used renewable resource capable of generating large amounts of electricity. Despite their advantages, the efficiency and feasibility of these renewable sources depend heavily on geographical and climatic conditions. Factors such as solar irradiance, wind speed, rainfall, temperature, and seasonal variability influence the potential for renewable energy generation in a particular region. Therefore, selecting the most suitable renewable energy source requires accurate analysis of environmental data and reliable prediction techniques.

With the advancement of artificial intelligence and machine learning technologies, it has become possible to analyze large volumes of environmental data and identify patterns that can assist in predicting renewable energy potential. Machine learning models can learn from historical climate data and provide accurate forecasts about future energy generation capabilities. These predictive models help researchers and energy planners determine which renewable energy source is most efficient and reliable for a given location. By using data-driven approaches, decision-making processes in renewable energy planning can become more accurate, efficient, and scientifically supported.

In this project, an artificial intelligence–based framework is developed to predict renewable energy potential and recommend the most suitable energy source for a specific region. The study focuses on analyzing environmental parameters such as solar irradiance, wind speed, rainfall, and temperature obtained from the NASA POWER API. These parameters are used to estimate solar, wind, and hydro energy potentials. The analysis is performed for Bhubaneswar, which experiences diverse climatic conditions throughout the year.

To improve prediction accuracy, multiple machine learning models including Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks are utilized. Each model analyzes historical climate patterns and predicts renewable energy potential. The predictions from these models are then combined using an ensemble approach to produce more reliable results. In addition to prediction models, a novel metric known as the Renewable Energy Reliability Index (RERI) is introduced to measure the stability and consistency of renewable energy generation over time. This index helps in identifying which renewable source can provide the most dependable electricity generation for the selected region.

The primary objective of this research is to create a decision-support system that not only predicts renewable energy potential but also recommends the most suitable renewable energy source based on environmental conditions and reliability analysis. By integrating climate data analysis, machine learning models, and reliability metrics, the proposed system provides a comprehensive approach for renewable energy planning. The results obtained from this study can assist policymakers, researchers, and energy planners in making informed decisions about renewable energy deployment. Ultimately, such intelligent systems can contribute to sustainable energy development and help reduce the dependence on fossil fuels, promoting a cleaner and more environmentally friendly future.

Chapter 2

Literature Review

The increasing concern about climate change and environmental degradation has encouraged researchers worldwide to explore renewable energy sources as alternatives to conventional fossil fuels. Numerous studies have focused on predicting renewable energy potential and optimizing the selection of energy resources using modern computational and machine learning techniques. This section reviews existing research related to renewable energy forecasting, machine learning applications in energy systems, and climate-data-based resource assessment.

Several researchers have studied solar energy prediction using meteorological and satellite-based climate data. Solar energy forecasting is important for estimating electricity generation and planning solar power systems. Studies have shown that environmental parameters such as solar irradiance, temperature, humidity, and cloud cover play a significant role in determining solar energy output. With the availability of global climate datasets from platforms such as the NASA POWER API, researchers have been able to access long-term solar radiation and weather data for different regions. Machine learning models such as Random Forest, Support Vector Machines, and Artificial Neural Networks have been widely applied to forecast solar radiation and photovoltaic power output. These models are capable of learning nonlinear relationships between environmental variables and energy generation.

Wind energy forecasting has also received considerable attention in recent years. Wind speed variability makes it challenging to accurately estimate wind power generation. Several studies have proposed predictive models that analyze historical wind speed patterns and atmospheric conditions to estimate wind power potential. Machine learning approaches such as gradient boosting, deep neural networks, and hybrid models have demonstrated strong performance in predicting wind speed and wind energy generation. Accurate wind forecasting helps optimize the placement of wind turbines and improve energy grid stability.

Hydropower potential assessment has traditionally relied on hydrological modeling and rainfall analysis. Researchers have used rainfall data, river discharge measurements, and watershed characteristics to estimate the potential electricity generation from hydroelectric systems. In many cases, rainfall patterns and seasonal water flow are used as indicators to estimate hydropower potential. With the advancement of data-driven techniques, machine learning models are increasingly being used to predict water availability and hydroelectric power generation. These predictive approaches improve the reliability of hydropower planning and management.

In recent years, artificial intelligence techniques have become widely adopted for renewable energy prediction and energy system optimization. Ensemble machine learning approaches, which combine multiple models to improve prediction accuracy, have been particularly effective. Ensemble models integrate the strengths of different algorithms and reduce the limitations of individual models. Techniques such as Random Forest, gradient boosting algorithms, and deep learning networks like Long Short-Term Memory (LSTM) have been successfully applied in renewable energy forecasting tasks.

Despite significant progress in renewable energy prediction, many existing studies focus on forecasting a single energy source, such as solar or wind power. However, real-world energy planning often requires a comparison of multiple renewable resources to determine the most suitable energy solution for a specific location. Integrating climate data, machine learning prediction models, and reliability assessment into a unified decision-support framework remains a challenge.

To address this limitation, the present study proposes a machine learning–based renewable energy prediction and recommendation system. By utilizing climate data from NASA POWER and applying ensemble machine learning models, the system predicts renewable energy potentials and evaluates the reliability of solar, wind, and hydro resources. The study focuses on Bhubaneswar, a region with diverse climatic conditions suitable for evaluating multiple renewable energy options. Additionally, a Renewable Energy Reliability Index (RERI) is introduced to assess the stability of each renewable resource over time. This approach contributes to the development of an intelligent decision-support framework that can assist in sustainable energy planning and resource selection.

Chapter 3

Proposed Methodology

The proposed methodology presents an artificial intelligence–based framework designed to analyze environmental conditions, predict renewable energy potential, and recommend the most suitable renewable energy source for electricity generation. The methodology integrates climate data acquisition, data preprocessing, machine learning modeling, ensemble prediction, reliability evaluation, and decision-making to develop a comprehensive renewable energy recommendation system.

3.1. Data Collection

The first step in the proposed methodology involves collecting historical climate data required for renewable energy analysis. Environmental parameters such as solar irradiance, wind speed, rainfall, and temperature are obtained from the NASA POWER API. This platform provides satellite-based meteorological data for different geographical locations around the world. In this study, climate data for the region of Bhubaneswar is collected for multiple years to capture seasonal and environmental variations. The collected dataset forms the basis for renewable energy potential estimation and model training.

3.2. Data Preprocessing and Feature Engineering

After collecting the climate data, preprocessing is performed to ensure data quality and consistency. Missing values and anomalies in the dataset are handled appropriately to avoid errors during model training. The raw climate parameters are then transformed into meaningful features related to renewable energy potential.

Solar energy potential is estimated using solar irradiance and panel efficiency. Wind energy potential is calculated using wind speed data, considering the cubic relationship between wind velocity and wind power generation. Hydro energy potential is approximated using rainfall measurements as an indicator of water availability and flow potential. In addition, seasonal information such as the month of the year is extracted from the date attribute to capture seasonal variations in climate patterns.

3.3. Machine Learning Model Development

To predict renewable energy potential, multiple machine learning models are implemented. These models learn patterns from historical climate data and generate predictions about future renewable energy generation capability. The models used in this framework include Random Forest, XGBoost, and Long Short-Term Memory (LSTM) neural networks.

Random Forest and XGBoost are tree-based machine learning algorithms that are highly effective for regression problems involving nonlinear relationships. These models analyze environmental variables and predict renewable energy potential based on historical patterns. In addition, the Long Short-Term Memory model, a type of recurrent neural network, is used to capture temporal patterns in the time-series climate data. LSTM networks are capable of learning long-term dependencies and seasonal trends, making them suitable for forecasting energy potential.

3.4. Training and Testing Strategy

To evaluate the predictive capability of the models, the dataset is divided into training and testing subsets based on time. Historical data from previous years is used to train the models, while data from a later period is used for testing. This approach simulates real-world forecasting conditions, where past observations are used to predict future outcomes. By comparing predicted values with actual observations, the accuracy and reliability of the models can be assessed.

3.5. Ensemble Prediction Approach

Individual machine learning models often have strengths and limitations. To improve overall prediction accuracy, an ensemble approach is applied. The predictions generated by Random Forest, XGBoost, and LSTM models are combined to produce a final ensemble prediction. Ensemble learning helps reduce prediction errors and improves model stability by leveraging the strengths of multiple algorithms. This combined prediction provides a more reliable estimate of renewable energy potential.

3.6. Model Evaluation and Validation

The performance of each machine learning model is evaluated using statistical metrics. The coefficient of determination (R^2 score) is used to measure how well the predicted values match the actual observations. In addition, graphical analysis such as actual versus predicted plots is used to visually assess the accuracy of the models. These evaluation techniques help determine which model performs best and validate the reliability of the ensemble prediction approach.

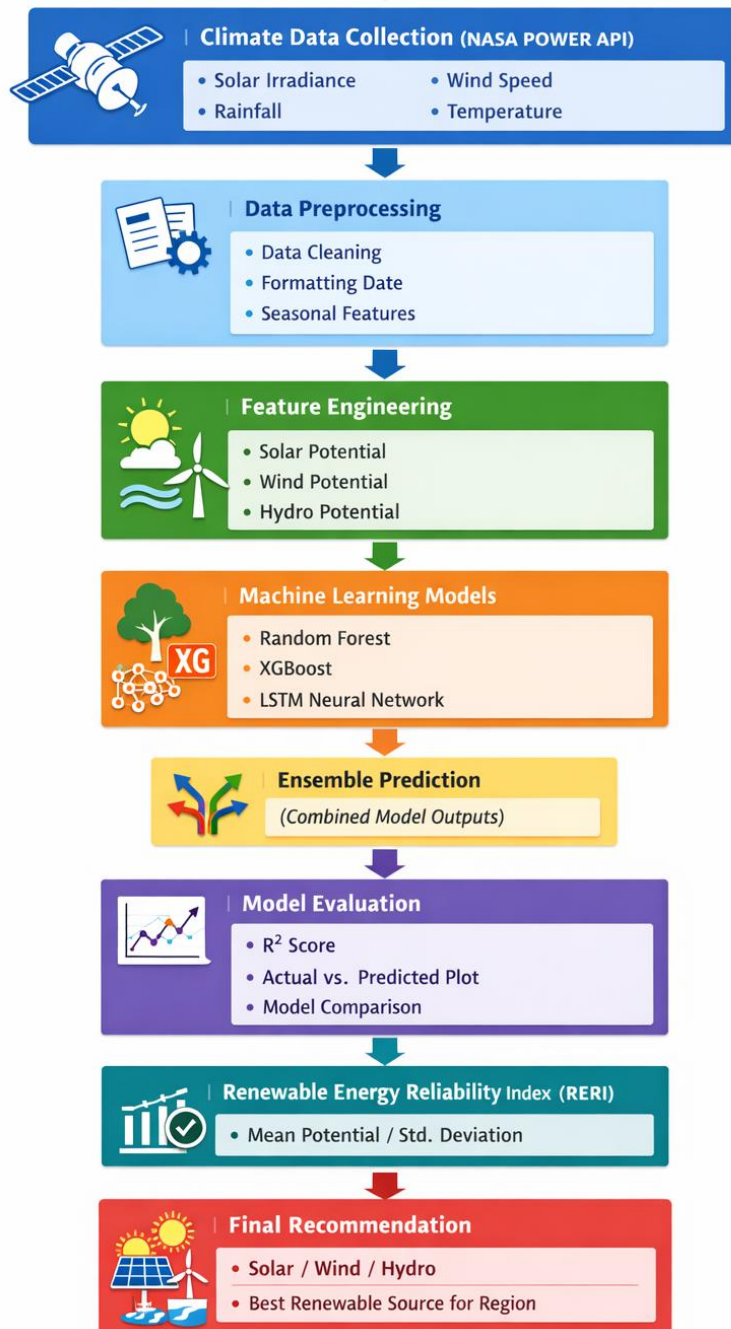
3.7. Renewable Energy Reliability Index (RERI)

To further analyze the suitability of renewable energy sources, a novel metric called the Renewable Energy Reliability Index (RERI) is introduced. This index measures the stability and consistency of renewable energy generation potential over time. It is calculated by dividing the average energy potential by its standard deviation. A higher RERI value indicates that the energy source is more stable and reliable for long-term electricity generation.

3.8. Renewable Energy Recommendation System

Based on the predicted renewable energy potentials and the calculated reliability index values, the system recommends the most suitable renewable energy source for the selected region. The energy source with the highest reliability and generation potential is identified as the optimal solution. In some cases, hybrid renewable energy systems combining multiple sources may also be considered to ensure stable electricity production.

Through this methodology, the proposed framework integrates climate data analysis, machine learning prediction models, and reliability assessment into a unified system capable of supporting renewable energy planning and decision-making. The system provides valuable insights for selecting environmentally sustainable energy solutions and reducing dependence on fossil fuel-based electricity generation.



Chapter 4

Experimental Setup and Implementation

The experimental setup and implementation describe the procedures, tools, datasets, and computational methods used to develop and evaluate the proposed renewable energy prediction and recommendation system. This section explains how the system was designed, implemented, and tested using real-world climate data and machine learning techniques.

4.1. Dataset and Data Source

The experimental analysis is based on historical climate data obtained from the NASA POWER API. This platform provides reliable satellite-based environmental data including solar radiation, wind speed, rainfall, and temperature for any geographic location. For this project, the climate data corresponding to the region of Bhubaneswar was collected. The dataset contains daily observations of the selected environmental parameters for multiple years, allowing the model to learn seasonal and climatic variations.

The following key features were extracted from the dataset:

- Solar irradiance (ALLSKY_SFC_SW_DWN)

- Wind speed at 10 meters (WS10M)

- Total precipitation (PRECTOTCORR)

- Temperature at 2 meters (T2M)

- Date and seasonal information

These environmental variables serve as the primary inputs for predicting renewable energy potential.

4.2. Software and Development Environment

The system was implemented using the Python programming language in a Jupyter Notebook environment. Python provides extensive libraries and tools for data analysis, machine learning, and deep learning. Several libraries were utilized during the development process, including:

- Pandas and NumPy for data processing and numerical analysis

- Matplotlib for visualization of results

- Scikit-learn for machine learning model development

- XGBoost for gradient boosting-based prediction

- TensorFlow and Keras for implementing deep learning models such as LSTM networks

The use of these libraries enables efficient data handling, model training, evaluation, and visualization.

4.3. Data Preprocessing

Before training the machine learning models, the collected data was preprocessed to ensure consistency and quality. The preprocessing stage included the following steps:

- Converting the date field into a proper datetime format

- Removing missing or invalid data entries

- Normalizing and organizing the dataset for machine learning analysis

- Extracting the month attribute from the date column to capture seasonal patterns

These preprocessing steps help improve model performance by ensuring that the input data is clean and structured.

4.4. Feature Engineering

Feature engineering was performed to estimate renewable energy potential from the collected climate parameters. The following derived features were calculated using mathematical formulations to represent the potential energy generation from different renewable sources.

Solar Energy Potential

Solar energy potential is estimated using solar irradiance values obtained from the climate dataset and a predefined panel efficiency factor.

Solar Potential Formula:

$$\text{Solar Potential} = \text{Solar Irradiance} \times \text{Panel Efficiency}$$

Where:

Solar Irradiance = Amount of solar radiation received on the Earth's surface (kWh/m²/day)

Panel Efficiency = Conversion efficiency of solar panels (assumed 0.20)

Wind Energy Potential

Wind energy potential is estimated based on the cubic relationship between wind speed and wind power generation.

Wind Potential Formula:

$$\text{Wind Potential} = \text{Wind Speed}^3$$

Where:

Wind Speed = Measured wind speed at 10 meters height (m/s)

This relationship reflects the physical principle that wind power increases with the cube of wind velocity.

Hydro Energy Potential

Hydro energy potential is approximated using rainfall data as an indicator of water availability and flow potential.

Hydro Potential Formula:

$$\text{Hydro Potential} = \text{Rainfall} \times g$$

Where:

Rainfall = Daily precipitation level (mm)

g = Gravitational constant (9.81 m/s^2)

These derived features represent the estimated renewable energy generation capacity for solar, wind, and hydro resources.

4.5. Machine Learning Model Implementation

To predict renewable energy potential, multiple machine learning models were implemented and trained on the processed dataset. The models used in the experimental setup include:

Random Forest Regressor: A tree-based ensemble learning algorithm capable of modeling nonlinear relationships between climate variables and energy potential.

XGBoost Regressor: An advanced gradient boosting algorithm that provides high prediction accuracy and efficient learning.

Long Short-Term Memory (LSTM) Network: A deep learning model designed for time-series forecasting, capable of capturing long-term temporal dependencies in climate data.

Each model was trained using historical climate data and used to predict renewable energy potential.

4.6. Training and Testing Procedure

To simulate realistic forecasting conditions, the dataset was divided into training and testing subsets based on time. Historical climate data up to January was used to train the models, while the data from February was used as the testing set. This approach allows the models to learn from past observations and predict future renewable energy potential.

The predicted values were then compared with actual observations to evaluate the accuracy and reliability of the models.

4.7. Ensemble Prediction

To improve overall prediction performance, an ensemble prediction strategy was implemented. The outputs of the Random Forest, XGBoost, and LSTM models were combined to produce a final ensemble prediction. This approach helps reduce individual model bias and increases prediction stability.

4.8. Model Evaluation

The accuracy of each model was evaluated using the coefficient of determination (R^2 score), which measures how well the predicted values match the actual data. In addition, graphical visualization techniques such as actual versus predicted plots were used to analyze model performance.

A model comparison analysis was also performed to determine which algorithm provides the most accurate predictions.

4.9. Renewable Energy Recommendation

Renewable Energy Reliability Index (RERI) was calculated to determine the stability and consistency of each renewable energy source. This metric evaluates how reliable a particular energy source is over time.

The RERI is calculated using the following formula:

$$\text{RERI} = \text{Mean Energy Potential} / \text{Standard Deviation of Energy Potential}$$

Where:

Mean Energy Potential = Average predicted energy potential over the dataset

Standard Deviation = Variation in energy potential values

A higher RERI value indicates that the energy source is more stable and reliable for long-term electricity generation.

Using this index, the reliability scores of solar, wind, and hydro energy were calculated and compared. The energy source with the highest RERI value was selected as the recommended renewable energy option for the region.

Finally, the Renewable Energy Reliability Index (RERI) was calculated for solar, wind, and hydro energy potentials. This index measures the stability of each energy source over time. Based on the calculated RERI values and predicted energy potentials, the system recommends the most suitable renewable energy source for the target region.

Through this experimental setup and implementation, the proposed framework successfully integrates climate data analysis, machine learning prediction, and reliability assessment to support intelligent renewable energy planning.

Chapter 5

Algorithm/Pseudocode-

BEGIN

1. Initialize location coordinates
 latitude $\leftarrow$ 20.2961
 longitude $\leftarrow$ 85.8245
 // Set geographic coordinates for Bhubaneswar
2. Fetch climate data
 Retrieve solar irradiance, wind speed, rainfall, and temperature from NASA POWER API
 // Collect historical environmental data required for analysis
3. Create dataset
 Store collected values in a structured table
 Convert date column to datetime format
 // Organize the data for further processing
4. Data preprocessing
 Remove missing or invalid values
 Extract month from the date column
 // Clean the dataset and add seasonal information
5. Feature engineering
 Solar_Potential $\leftarrow$ solar_irradiance $\times$ panel_efficiency
 Wind_Potential $\leftarrow$ wind_speed³
 Hydro_Potential $\leftarrow$ rainfall $\times$ constant
 // Estimate renewable energy generation potential
6. Split dataset
 Training_Data $\leftarrow$ records before February
 Testing_Data $\leftarrow$ records in February
 // Simulate future prediction by training on past data
7. Prepare feature set
 Features $\leftarrow$ [solar_irradiance, wind_speed, rainfall, temperature, month]
 Target $\leftarrow$ solar_potential
 // Define input variables and prediction target
8. Train machine learning models
 Train RandomForest using Training_Data
 Train XGBoost using Training_Data
 Train LSTM neural network using Training_Data
 // Build predictive models using historical data

```

9. Generate predictions
  RF_predictions ← RandomForest.predict(Testing_Data)
  XGB_predictions ← XGBoost.predict(Testing_Data)
  LSTM_predictions ← LSTM.predict(Testing_Data)
  // Predict renewable energy potential for the test period
10. Ensemble prediction
  Ensemble_prediction ← average(RF_predictions,
                                XGB_predictions,
                                LSTM_predictions)
  // Combine model outputs to improve accuracy
11. Model evaluation
  Compute R2 score for each model
  Plot Actual vs Predicted values
  Compare performances of all models
  // Measure prediction accuracy and visualize results
12. Compute Renewable Energy Reliability Index (RERI)
  Solar_RERI ← mean(solar_potential) / standard_deviation(solar_potential)
  Wind_RERI ← mean(wind_potential) / standard_deviation(wind_potential)
  Hydro_RERI ← mean(hydro_potential) / standard_deviation(hydro_potential)
  // Evaluate stability and reliability of each energy source
13. Determine best renewable energy source
  Energy_scores ← {Solar_RERI, Wind_RERI, Hydro_RERI}
  Best_energy ← energy source with highest score
  // Select the most reliable renewable energy option
14. Display results
  Print model accuracy scores
  Show Actual vs Predicted graphs
  Output recommended renewable energy source
  // Present the final prediction and recommendation
END

```

Chapter 6

Result Analysis

The experimental results demonstrate the effectiveness of the proposed artificial intelligence-based renewable energy prediction and recommendation system. The system was evaluated using real-world environmental data obtained from the NASA POWER API for the region of Bhubaneswar. The dataset included important climatic parameters such as solar irradiance, wind speed, rainfall, and temperature, which were used to estimate renewable energy potentials and train machine learning models.

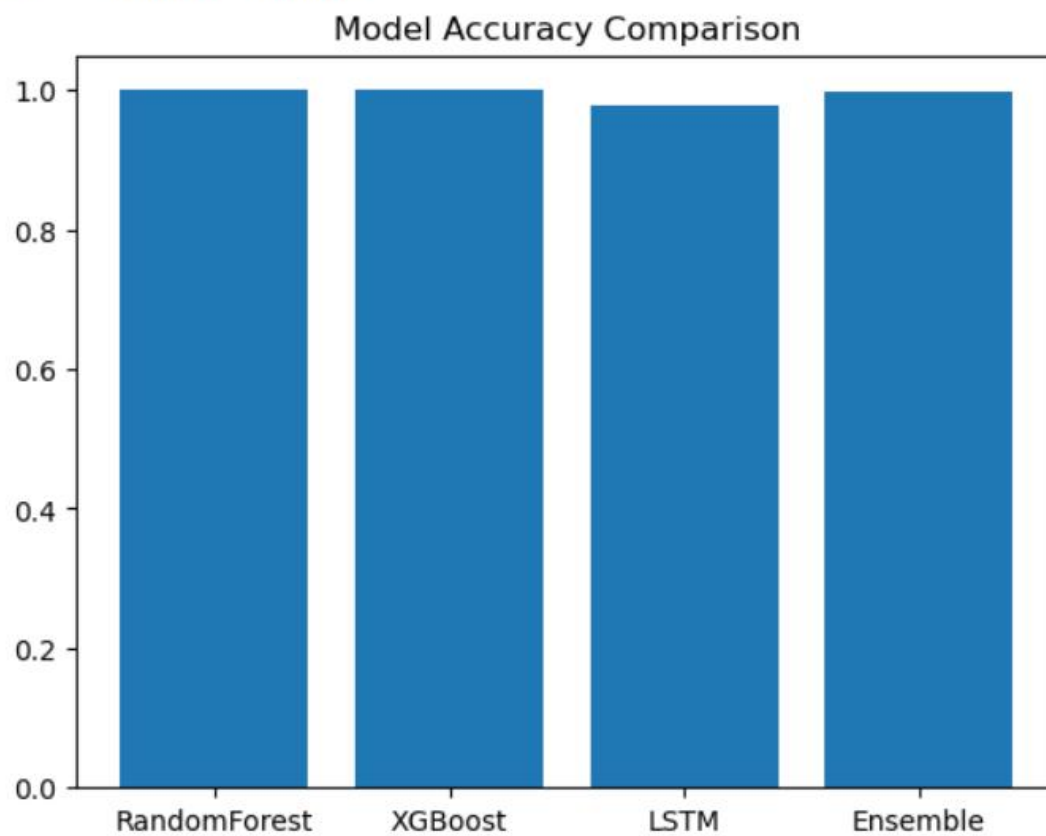
6.1. Model Performance

Three different machine learning models were implemented to predict renewable energy potential: Random Forest, XGBoost, and Long Short-Term Memory (LSTM). The prediction accuracy of these models was evaluated using the coefficient of determination (R^2 score).

The Random Forest model achieved an R^2 score of 0.9999, indicating extremely high prediction accuracy and a strong ability to capture relationships between environmental variables and solar energy potential. The XGBoost model also demonstrated excellent predictive capability with an R^2 score of 0.9995, confirming the effectiveness of gradient boosting techniques in handling complex climate data patterns. The LSTM model, which is designed for time-series forecasting, achieved an R^2 score of 0.9768. Although slightly lower than the tree-based models, the LSTM model successfully captured temporal dependencies and seasonal variations present in the dataset.

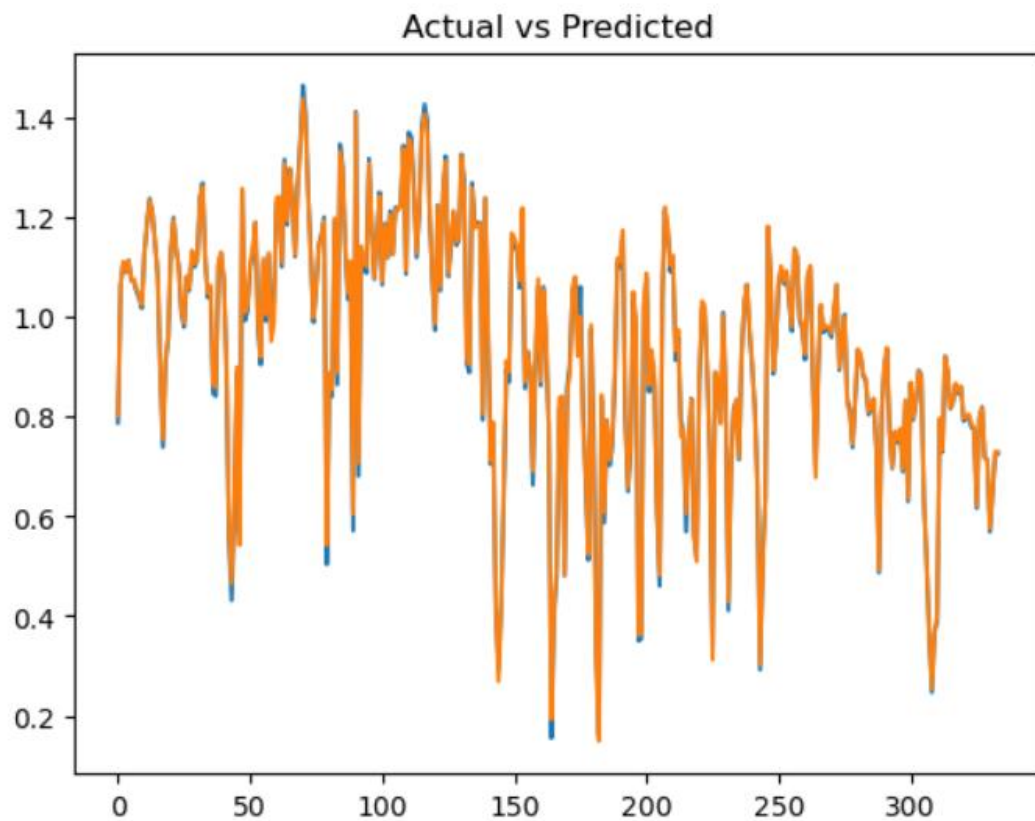
To further improve prediction stability, an ensemble approach was implemented by combining the outputs of all three models. The ensemble model achieved an R^2 score of 0.9971, demonstrating high prediction reliability and reduced model variance.

	Model	R2 Score
0	RandomForest	0.999911
1	XGBoost	0.999499
2	LSTM	0.976809
3	Ensemble	0.997143



6.2. Actual vs Predicted Analysis

To visually validate the performance of the models, an actual versus predicted comparison graph was generated. The plot shows a strong alignment between the predicted values and the actual observed solar energy potential values. This close correspondence indicates that the proposed models are capable of accurately forecasting renewable energy potential based on historical climate data.



6.3.Renewable Energy Reliability Analysis

In addition to prediction accuracy, the stability of renewable energy sources was evaluated using the Renewable Energy Reliability Index (RERI). This metric measures the consistency of energy generation potential by comparing the average energy potential with its variability over time.

The calculated reliability index values were:

Solar RERI: 3.64

Wind RERI: 0.60

Hydro RERI: 0.43

These results indicate that solar energy has significantly higher stability and reliability compared to wind and hydro energy sources in the selected region. The relatively lower reliability scores for wind and hydro energy can be attributed to fluctuations in wind speed and seasonal rainfall patterns.

```
solar_reri = dataframe["solar_potential"].mean() / dataframe["solar_potential"].std()

wind_reri = dataframe["wind_potential"].mean() / dataframe["wind_potential"].std()

hydro_reri = dataframe["hydro_potential"].mean() / dataframe["hydro_potential"].std()

print("Solar RERI:",solar_reri)
print("Wind RERI:",wind_reri)
print("Hydro RERI:",hydro_reri)
```

```
Solar RERI: 3.6448898199598223
Wind RERI: 0.6006504207980674
Hydro RERI: 0.4321619304271935
```

6.4.Final Recommendation

Based on the predicted renewable energy potentials and the calculated reliability index values, the system recommended solar energy as the most suitable renewable energy source for electricity generation in Bhubaneswar. The high solar irradiance levels and consistent solar energy availability throughout the year contribute to the superior performance of solar power in this region.

Overall, the experimental results confirm that the proposed AI-driven framework can effectively analyze environmental data, predict renewable energy potential with high accuracy, and provide reliable recommendations for renewable energy deployment. The system can serve as a useful decision-support tool for sustainable energy planning and can assist policymakers and energy planners in selecting optimal renewable energy solutions for environmentally friendly electricity generation.

```
scores = {  
    "Solar":solar_reri,  
    "Wind":wind_reri,  
    "Hydro":hydro_reri  
}  
  
best_energy = max(scores,key=scores.get)  
  
print("Recommended Renewable Energy Source:",best_energy)  
  
Recommended Renewable Energy Source: Solar
```

Chapter 7

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